Nimish Agarwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.12
Koncept Public School, Chakradharpur	Class XII (CBSE)	2024	92.60%
St. Xavier's English School, Chakradharpur	Class X (ICSE)	2022	97.80%

SCHOLASTIC AND TECHNICAL ACHIEVEMENTS

- **Candidate Master** on Codeforces with a peak rating of **1903** (Top **250** in India), having solved over **1000** problems on the platform. (Profile)
- Secured **Rank 82** in **IICPC CodeFest 2026 Regionals**, after qualifying for **IICPC CodeFest 2026 Prelims** with **Rank 138**.
- Secured **Rank 359** in **IICPC QuantFest 2025 Prelims**, qualifying for the **IICPC QuantFest 2025 Regionals**.
- Secured **Rank 4** in the **Integral Cup 2026 Qualifier**.

SKILLS

- **Languages:** C++, C, Python, Java, OCaml
- **Familiar with:** Bash, NumPy, Pandas, Matplotlib, SQL
- **Tools:** Git, GitHub, Docker, PostgreSQL, LaTeX
- **Soft Skills:** Communication, Work Ethic, Teamwork, Problem Solving, Time Management

PROJECTS

xv6 OS Kernel Extensions

(Part of Operating Systems Coursework) | *GitHub*

- Replaced the xv6 scheduler with a 4-level **syscall-aware MLFQ**, improving responsiveness while preventing starvation.
- Implemented **Clock page replacement** with page fault handling and virtual memory management.
- Developed a **disk-backed swap system** with **FCFS/SSTF** disk scheduling and **RAID 0/1/5** storage abstractions.

Advanced Pipelined RISC-V Processor Simulator

(Part of Computer Architecture Coursework) | *GitHub*

- Developed a cycle-accurate **5-stage pipelined** RISC-V processor simulator with complete pipeline execution.
- Implemented a **Hazard Management Unit** featuring data forwarding, stalling, and **BTFNT** and **1-bit dynamic branch prediction** to resolve pipeline hazards and reduce misprediction penalties.
- Designed a configurable **L1 cache** supporting variable associativity, replacement policies, and write strategies.

Core-ML Exception Handling Extension

(Part of Software Fundamentals Coursework) | *GitHub*

- Extended the Core-ML grammar, AST, and semantics with **try-catch** exception handling and **raise** expressions.
- Implemented **IRaise** and **IHandle** across the type checker, compiler, and virtual machine for exception propagation.

- Added runtime support for arithmetic exceptions, including **DivisionByZero**, with comprehensive testing.

Infinite Precision Arithmetic calculator

(Part of Software Fundamentals Coursework) | *GitHub*

- Designed **AInteger** and **AFloat** classes for arbitrary-precision integer and floating-point arithmetic.
- Implemented reusable arithmetic, parsing, comparison, and signed operations **without** built-in big number libraries.
- Developed algorithms for normalization, decimal alignment, and quotient/remainder computation.

Statistical Data Analysis

(Part of Applied Statistics coursework) | *Link to reports*

- Performed statistical inference using sampling, point/interval estimation, confidence intervals, and hypothesis testing.
- Applied t-tests, Chi-square tests, MLE, MOM, and correlation analysis using Python on real-world datasets.
- Visualized and modeled data with Matplotlib, validating the Central Limit Theorem and fitting Gamma distributions.

RELEVANT COURSES

- **Mathematics:** Discrete Mathematics, Linear Algebra, Probability & Random Variables, Applied Statistics, Calculus I & II, Transform Techniques, Introduction to Statistics

- **Computer Science:** Algorithms, Data Structures, Operating Systems I & II, Database Management Systems I & II, Computer Architecture, Theory of Computation, Introduction to Programming, Introduction to AI, Compilers I, Software Development Fundamentals

POSITIONS OF RESPONSIBILITY

Overall Head, Infero

Algorithmic and Logical Reasoning Club, IIT Hyderabad

- Lead and coordinate all club activities while managing the club's core team.
- Organize and conduct internship and placement-oriented coding contests for students.

Core Member, Infero

Algorithmic and Logical Reasoning Club, IIT Hyderabad

- Mentored juniors in competitive programming through training sessions and contests.
- Contributed to problem setting for **Elan & Nvision CodeClash 2025**.

Teaching Assistant, *Discete Mathematics (CS1010)*

Internship Coordinator, Office of Career Services